

COUNTING FAITHFUL FAMILIES OF INTEGRALS IN WEAK ALGEBRAIC QUANTUM GROUPS

ABSTRACT. We introduce an invariant $N(A)$, defined as the minimal size of a faithful family of left or right integrals for a regular weak multiplier Hopf algebra with sufficiently many integrals. Using the Van Daele–Wang spanning criterion, we characterise $N(A)$ and prove $N(A) \leq 2 \dim \varepsilon_t(A)$ when the target algebra $\varepsilon_t(A)$ is finite-dimensional. In the $*$ -algebraic setting, we show that for positive left integrals, either spanning condition from Van Daele–Wang alone ensures faithfulness, and any finite faithful family of positive left integrals can be merged into one faithful positive integral. Hence, if $\varepsilon_t(A)$ is finite-dimensional and there is a faithful family of positive left integrals, then $N(A) = 1$. We compute $N(A)$ for groupoid examples and always find $N(A) = 1$. For packed algebras from finite-object Hopf categories, the object count gives only a coarse upper bound, but in the finite-dimensional Frobenius case we again obtain $N(A) = 1$. Collectively, these results provide a partial answer to a question posed by Van Daele and Wang.

Keywords: weak multiplier Hopf algebra; weak algebraic quantum group; integral; groupoid; Hopf category; Frobenius algebra

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1. INTRODUCTION

Weak multiplier Hopf algebras (also called **weak algebraic quantum groups**) provide a natural common framework for Hopf algebras, weak Hopf algebras, and multiplier Hopf algebras. Their theory was developed by Van Daele and Wang in a series of papers [11, 12, 13], and, in the regular case, this theory supports a rich integral theory and a duality theory. Most recently, Van Daele and Wang established duality for regular weak multiplier Hopf algebras with sufficiently many integrals [14]. These developments belong to a broader algebraic approach to quantum symmetry, going back to Van Daele’s algebraic framework for group duality [10] and continuing through Timmermann’s work on algebraic quantum groupoids [8] and the later pairing and duality theory of Timmermann, Van Daele, and Wang [9].

Recent work has expanded this picture in several directions. On the one hand, Wang and collaborators have developed further aspects of the subject, including pairing and duality of algebraic quantum groupoids [9], Drinfeld doubles for infinite groupoid examples [15], and groupoid-cograded and quasitriangular weak multiplier Hopf algebras [5, 6]. On the other hand, related developments by other authors include partial $*$ -algebraic quantum groups [4] and Hopf categories together with their packed weak (multiplier) Hopf algebras [1, 2]. These works provide both new constructions and a growing supply of examples.

A striking feature of the integral theory is that, unlike the multiplier Hopf algebra case, a non-zero integral need not be faithful and need not be unique up to scalar. In general, one is therefore forced to work with faithful families of integrals rather than with a single distinguished integral. This leads to a natural numerical problem, already highlighted in the discussion of Van Daele and Wang [14]: how many integrals are actually needed to obtain a faithful family, when is this number finite, and how does it behave in the basic examples coming from groupoids and Hopf categories?

In this paper we isolate this quantity as an invariant and study it directly. For a weak algebraic quantum group A , we define $N_l(A)$ and $N_r(A)$ to be the minimal cardinalities of faithful families of left and right integrals, prove that they coincide, and reinterpret the common value $N(A)$ through the Van Daele–Wang spanning criterion. We then prove that

$$N(A) \leq 2 \dim \varepsilon_t(A)$$

whenever the target algebra is finite-dimensional. In the $*$ -algebraic setting, we show that for positive left integrals either of the two spanning conditions already implies faithfulness, and that every finite faithful family of positive left integrals sums to a single faithful positive integral. As a consequence, finite-dimensional target algebra together with a faithful family of positive left integrals forces $N(A) = 1$. We also compute $N(A)$ in the groupoid examples, where it is always equal to 1, and study packed algebras of finite-object Hopf categories, where the number of objects yields only a coarse upper bound while the finite-dimensional Frobenius case again gives $N(A) = 1$. To the best of our knowledge, this is the first paper to isolate and study this numerical aspect of the theory in its own right.

The paper is organised as follows. In Section 2 we recall the faithful-family criterion for integrals and introduce the invariants $N_l(A)$, $N_r(A)$, and

$N(A)$. In Section 3 we prove the general finite-dimensional bound and the collapse results for positive integrals in the $*$ -setting. In Section 4 we compute the invariant in the groupoid examples and analyse the corresponding bounds and collapse phenomena for packed algebras of Hopf categories.

2. WEAK ALGEBRAIC QUANTUM GROUPS AND THE COUNTING PROBLEM

We work over the ground field $\mathbb{C}$. Throughout, (A, Δ) denotes a regular weak multiplier Hopf algebra in the sense of [12]. We write $E \in M(A \otimes A)$ for the canonical idempotent, S for the antipode, and

$$\varepsilon_s, \varepsilon_t : A \rightarrow M(A)$$

for the source and target maps. Unless stated otherwise, we follow the notation of [14].

Definition 2.1. Define

$$\mathcal{I}_l(A) := \{\phi : A \rightarrow \mathbb{C} \text{ linear} : (\iota \otimes \phi)\Delta(a) \in \varepsilon_t(A) \text{ for all } a \in A\},$$

and

$$\mathcal{I}_r(A) := \{\psi : A \rightarrow \mathbb{C} \text{ linear} : (\psi \otimes \iota)\Delta(a) \in \varepsilon_s(A) \text{ for all } a \in A\}.$$

A non-zero element of $\mathcal{I}_l(A)$ is called a *left integral*, and a non-zero element of $\mathcal{I}_r(A)$ is called a *right integral*.

A family $\Phi \subseteq \mathcal{I}_l(A) \setminus \{0\}$ is called *faithful* if, for every $x \in A$,

$$\phi(xa) = 0 \text{ for all } a \in A \text{ and } \phi \in \Phi \implies x = 0,$$

and

$$\phi(ax) = 0 \text{ for all } a \in A \text{ and } \phi \in \Phi \implies x = 0.$$

If (A, Δ) admits a faithful family of left integrals, we call it a *weak algebraic quantum group*.

For each $\phi \in \mathcal{I}_l(A)$ define

$$U_\phi^- := \{(\phi \otimes \iota)((a \otimes 1)E) : a \in A\} \subseteq \varepsilon_t(A),$$

$$U_\phi^+ := \{(\phi \otimes \iota)(E(a \otimes 1)) : a \in A\} \subseteq \varepsilon_t(A).$$

Standard properties of the canonical idempotent imply that these sets are contained in $\varepsilon_t(A)$, and linearity shows that they are vector subspaces.

The following result is a direct reformulation of the spanning criterion of Van Daele and Wang.

Proposition 2.2. Let $\Phi \subseteq \mathcal{I}_l(A) \setminus \{0\}$. Then Φ is faithful if and only if

$$\sum_{\phi \in \Phi} U_{\phi}^{-} = \varepsilon_t(A) \quad \text{and} \quad \sum_{\phi \in \Phi} U_{\phi}^{+} = \varepsilon_t(A).$$

Dually, a family $\Psi \subseteq \mathcal{I}_r(A) \setminus \{0\}$ is faithful if and only if the corresponding two families of subspaces span $\varepsilon_s(A)$.

Proposition 2.2 suggests the following natural numerical invariant.

Definition 2.3. Assume that (A, Δ) is a weak algebraic quantum group. Define

$$N_l(A) := \min\{|\Phi| : \Phi \subseteq \mathcal{I}_l(A) \setminus \{0\} \text{ is faithful}\},$$

and

$$N_r(A) := \min\{|\Psi| : \Psi \subseteq \mathcal{I}_r(A) \setminus \{0\} \text{ is faithful}\}.$$

These minima are well defined because faithful families exist by assumption.

Proposition 2.4. For every weak algebraic quantum group (A, Δ) one has

$$N_l(A) = N_r(A).$$

Proof. Let $\Phi \subseteq \mathcal{I}_l(A) \setminus \{0\}$ be faithful and set

$$\Phi^S := \{\phi \circ S : \phi \in \Phi\} \subseteq \mathcal{I}_r(A) \setminus \{0\}.$$

The antipode sends left integrals to right integrals, so Φ^S is well defined. Since S is bijective, we also have $|\Phi^S| = |\Phi|$. We claim that Φ^S is faithful.

Suppose first that $x \in A$ satisfies

$$(\phi \circ S)(xa) = 0 \quad \text{for all } a \in A \text{ and } \phi \in \Phi.$$

Then

$$\phi(S(a)S(x)) = 0 \quad \text{for all } a \in A \text{ and } \phi \in \Phi.$$

As S is surjective, this implies

$$\phi(bS(x)) = 0 \quad \text{for all } b \in A \text{ and } \phi \in \Phi.$$

Since Φ is faithful, we obtain $S(x) = 0$, hence $x = 0$.

Now assume that

$$(\phi \circ S)(ax) = 0 \quad \text{for all } a \in A \text{ and } \phi \in \Phi.$$

Then

$$\phi(S(x)S(a)) = 0 \quad \text{for all } a \in A \text{ and } \phi \in \Phi,$$

and again the surjectivity of S gives

$$\phi(S(x)b) = 0 \quad \text{for all } b \in A \text{ and } \phi \in \Phi.$$

Thus $S(x) = 0$, so $x = 0$. Hence Φ^S is faithful.

Therefore $N_r(A) \leq N_l(A)$. Applying the same argument to S^{-1} gives the reverse inequality. $\square$

Definition 2.5. In view of Proposition 2.4, we write

$$N(A) := N_l(A) = N_r(A)$$

and call $N(A)$ the *integral number* of (A, Δ) .

Equivalently,

$$N(A) = \min \left\{ |\Phi| : \Phi \subseteq \mathcal{I}_l(A) \setminus \{0\}, \sum_{\phi \in \Phi} U_\phi^- = \varepsilon_t(A) = \sum_{\phi \in \Phi} U_\phi^+ \right\}.$$

3. GENERAL BOUNDS AND THE POSITIVE *-CASE

We now prove the basic structural results for the invariant $N(A)$.

3.1. A general finite-dimensional bound. The first lemma is the family version of one half of the proof of [14, Lemma 2.16].

Lemma 3.1. *Let $\Phi \subseteq \mathcal{I}_l(A)$.*

(1) *If $\sum_{\phi \in \Phi} U_\phi^- = \varepsilon_t(A)$ and $b \in A$ satisfies*

$$\phi(ab) = 0 \quad \text{for all } a \in A \text{ and } \phi \in \Phi,$$

then $b = 0$.

(2) *If $\sum_{\phi \in \Phi} U_\phi^+ = \varepsilon_t(A)$ and $b \in A$ satisfies*

$$\phi(ba) = 0 \quad \text{for all } a \in A \text{ and } \phi \in \Phi,$$

then $b = 0$.

Proof. We prove (1). Fix $\phi \in \Phi$. The argument in the proof of [14, Lemma 2.16] shows that the assumption

$$\phi(ab) = 0 \quad \text{for all } a \in A$$

forces

$$b S^{-1}(x) = 0 \quad \text{for every } x \in U_\phi^-.$$

Since $\sum_{\phi \in \Phi} U_\phi^- = \varepsilon_t(A)$ and the antipode restricts to a bijection $S^{-1} : \varepsilon_t(A) \rightarrow \varepsilon_s(A)$, we conclude that

$$by = 0 \quad \text{for all } y \in \varepsilon_s(A).$$

By the non-degeneracy property of the source algebra recalled in [14, Section 1.2], this implies $b = 0$.

The proof of (2) is identical, using the second half of the proof of [14, Lemma 2.16]. $\square$

Theorem 3.2. *Let (A, Δ) be a weak algebraic quantum group. If*

$$\dim \varepsilon_t(A) = m < \infty,$$

then

$$N(A) \leq 2m.$$

In particular, every faithful family of left integrals contains a finite faithful subfamily.

Proof. Let $\Phi \subseteq \mathcal{I}_l(A)$ be a faithful family. By Proposition 2.2,

$$\sum_{\phi \in \Phi} U_\phi^- = \varepsilon_t(A) = \sum_{\phi \in \Phi} U_\phi^+.$$

As $\varepsilon_t(A)$ is m -dimensional, there exist $\phi_1^-, \dots, \phi_r^- \in \Phi$ with $r \leq m$ such that

$$\varepsilon_t(A) = U_{\phi_1^-}^- + \dots + U_{\phi_r^-}^-.$$

Similarly, there exist $\phi_1^+, \dots, \phi_s^+ \in \Phi$ with $s \leq m$ such that

$$\varepsilon_t(A) = U_{\phi_1^+}^+ + \dots + U_{\phi_s^+}^+.$$

Set

$$\Phi_0 := \{\phi_1^-, \dots, \phi_r^-, \phi_1^+, \dots, \phi_s^+\}.$$

Then $|\Phi_0| \leq r + s \leq 2m$, and again

$$\sum_{\phi \in \Phi_0} U_\phi^- = \varepsilon_t(A) = \sum_{\phi \in \Phi_0} U_\phi^+.$$

Hence Φ_0 is faithful by Proposition 2.2. The inequality follows. $\square$

Remark 3.3. Theorem 3.2 shows that the finite versus infinite dichotomy from [14, Problem 2, p. 66] can only become genuinely subtle when the

target algebra is infinite-dimensional. For all finite-dimensional weak Hopf algebra situations, the least faithful family is automatically finite.

3.2. Positive integrals on weak multiplier Hopf $*$ -algebras. We now assume that (A, Δ) is a regular weak multiplier Hopf $*$ -algebra. Recall from [14, Definition 2.1] that a left integral ϕ is **positive** when $\phi(a^*a) \geq 0$ for all $a \in A$.

Theorem 3.4. *Let $\Phi \subseteq \mathcal{I}_l(A)$ be a family of positive left integrals. Then the following are equivalent:*

- (1) Φ is faithful;
- (2) $\sum_{\phi \in \Phi} U_\phi^- = \varepsilon_t(A)$;
- (3) $\sum_{\phi \in \Phi} U_\phi^+ = \varepsilon_t(A)$.

Proof. The implication (1) $\Rightarrow$ (2), (3) is Proposition 2.2. Assume now that (2) holds.

By Lemma 3.1(1), if $b \in A$ satisfies $\phi(ab) = 0$ for all $a \in A$ and all $\phi \in \Phi$, then $b = 0$. It remains to treat the other kernel. So assume that

$$\phi(ba) = 0 \quad \text{for all } a \in A \text{ and } \phi \in \Phi.$$

Taking $a = b^*$ we get

$$\phi(bb^*) = 0 \quad \text{for all } \phi \in \Phi.$$

Since each ϕ is positive, the Cauchy–Schwarz inequality gives, for every $c \in A$,

$$|\phi(cb^*)|^2 \leq \phi(cc^*)\phi(bb^*) = 0.$$

Hence $\phi(cb^*) = 0$ for all $c \in A$ and all $\phi \in \Phi$. By the first part already proved, applied to b^* , we obtain $b^* = 0$, and therefore $b = 0$. Thus Φ is faithful.

The implication (3) $\Rightarrow$ (1) is proved in the same way, using Lemma 3.1(2). □

Theorem 3.5. *Let $\phi_1, \dots, \phi_n$ be positive left integrals on a regular weak multiplier Hopf $*$ -algebra. If $\{\phi_1, \dots, \phi_n\}$ is a faithful family, then*

$$\phi := \phi_1 + \dots + \phi_n$$

is a single faithful positive left integral. In particular, the existence of a finite faithful family of positive left integrals implies $N(A) = 1$.

Proof. The functional ϕ is again a positive left integral. We prove that it is faithful.

Assume first that

$$\phi(xa) = 0 \quad \text{for all } a \in A.$$

Putting $a = x^*$ yields

$$0 = \phi(xx^*) = \sum_{i=1}^n \phi_i(xx^*).$$

Each summand is non-negative, hence $\phi_i(xx^*) = 0$ for every i . Therefore, for every $a \in A$,

$$|\phi_i(xa)|^2 \leq \phi_i(xx^*)\phi_i(a^*a) = 0,$$

so $\phi_i(xa) = 0$ for all a and all i . Since the family $\{\phi_1, \dots, \phi_n\}$ is faithful, we get $x = 0$.

Now assume that

$$\phi(ax) = 0 \quad \text{for all } a \in A.$$

Taking $a = x^*$ gives

$$0 = \phi(x^*x) = \sum_{i=1}^n \phi_i(x^*x),$$

whence $\phi_i(x^*x) = 0$ for every i . Then

$$|\phi_i(ax)|^2 \leq \phi_i(aa^*)\phi_i(x^*x) = 0,$$

for every $a \in A$, so again $\phi_i(ax) = 0$ for all a and all i . Faithfulness of the family implies $x = 0$.

Thus ϕ is faithful. □

Combining Theorem 3.2 and Theorem 3.5 we obtain the following strong corollary.

Corollary 3.6. *Let (A, Δ) be a regular weak multiplier Hopf $*$ -algebra. Assume that $\dim \varepsilon_t(A) < \infty$ and that A admits a faithful family of positive left integrals. Then*

$$N(A) = 1.$$

Proof. By Theorem 3.2 there exists a finite faithful subfamily of positive left integrals. Theorem 3.5 then yields a single faithful positive left integral. □

Remark 3.7. Corollary 3.6 shows that, in the positive $*$ -setting, there is no genuinely finite obstruction. Any counterexample to $N(A) = 1$ must force every faithful positive family to be infinite, and hence must necessarily have infinite-dimensional target algebra.

4. GROUPOIDS AND HOPF CATEGORIES

We now compute $N(A)$ for the basic groupoid examples and examine the corresponding behaviour for packed Hopf categories.

4.1. Groupoids. Let G be a groupoid. We write $K(G)$ for the algebra of finitely supported complex-valued functions on G , with pointwise product and the coproduct of [11, Example 1.15], and $\mathbb{C}G$ for the groupoid algebra with the coproduct of [11, Example 1.16].

Theorem 4.1. *For every groupoid G one has*

$$N(K(G)) = 1, \quad N(\mathbb{C}G) = 1.$$

Proof. We first consider $K(G)$. As explained in the groupoid discussion of [14], every left integral on $K(G)$ is of the form

$$\phi_g(f) = \sum_{u \in G} g(u)f(u),$$

where g belongs to the multiplier source algebra, equivalently, g is constant on source fibres. In particular, taking $g = 1$ yields a left integral

$$\phi(f) = \sum_{u \in G} f(u).$$

For each $p \in G$, let $\delta_p \in K(G)$ be the characteristic function of $\{p\}$. If $f \in K(G)$ satisfies $\phi(fh) = 0$ for all $h \in K(G)$, then

$$0 = \phi(f\delta_p) = f(p)$$

for every $p \in G$. Hence $f = 0$. Since the product on $K(G)$ is commutative, the same argument applies to the condition $\phi(hf) = 0$ for all $h \in K(G)$. Thus ϕ is faithful, and therefore $N(K(G)) = 1$.

Now consider $\mathbb{C}G$, with canonical basis $\{\lambda_p : p \in G\}$. Again by the groupoid analysis in [14], the functional ψ defined by

$$\psi(\lambda_e) = 1 \quad (e \text{ a unit}), \quad \psi(\lambda_p) = 0 \quad (p \text{ not a unit})$$

is an integral. We verify directly that it is faithful. Write

$$b = \sum_{p \in G} c_p \lambda_p \in \mathbb{C}G.$$

Assume first that $\psi(\lambda_q b) = 0$ for all $q \in G$. Fix $p \in G$ and choose $q = p^{-1}$. Then

$$\lambda_{p^{-1}} b = \sum_{r \in G} c_r \lambda_{p^{-1}} \lambda_r.$$

The term $\lambda_{p^{-1}} \lambda_r$ is a unit if and only if $r = p$, in which case it equals $\lambda_{p^{-1}p}$. All remaining terms are either zero or basis vectors indexed by non-units, and are therefore annihilated by ψ . Hence

$$0 = \psi(\lambda_{p^{-1}} b) = c_p$$

for every $p \in G$, so $b = 0$. The argument for the condition $\psi(b \lambda_q) = 0$ for all $q \in G$ is identical. Therefore ψ is faithful, and so $N(\mathbb{C}G) = 1$. $\square$

Remark 4.2. Theorem 4.1 shows that neither the cardinality of the object set nor the number of connected components, by itself, controls $N(A)$: both may be infinite while the integral number is still equal to 1.

4.2. Packed Hopf categories. Let $\mathcal{H}$ be a $\mathbb{C}$ -linear Hopf category with object set X . Following [1], write

$$\widehat{\mathcal{H}} := \bigoplus_{x, y \in X} \mathcal{H}_{x, y}$$

for its packed algebra.

Proposition 4.3. Assume that X is finite and that $\widehat{\mathcal{H}}$ is a weak algebraic quantum group. Then

$$N(\widehat{\mathcal{H}}) \leq 2|X|.$$

Proof. By [1, Proposition 6.1], $\widehat{\mathcal{H}}$ is a weak Hopf algebra. Moreover, for $h \in \mathcal{H}_{x, y}$, the formula for the target map in the proof of [1, Proposition 6.1] gives

$$\varepsilon_t(h) = \varepsilon_{x, y}(h) 1_x.$$

It follows that

$$\varepsilon_t(\widehat{\mathcal{H}}) = \bigoplus_{x \in X} \mathbb{C} 1_x,$$

and therefore

$$\dim \varepsilon_t(\widehat{\mathcal{H}}) = |X|.$$

The claim now follows from Theorem 3.2. $\square$

Proposition 4.3 shows that the number of objects gives, at best, a coarse upper bound. In the finite-dimensional Frobenius situation the invariant again collapses to 1.

Theorem 4.4. *Let $\mathcal{H}$ be a $\mathbb{C}$ -linear Hopf category with finite object set X , and assume that each hom-space $\mathcal{H}_{x,y}$ is finite-dimensional. Then*

$$N(\widehat{\mathcal{H}}) = 1.$$

Proof. Since X is finite and all hom-spaces are finite-dimensional, the packed algebra $\widehat{\mathcal{H}}$ is a finite-dimensional weak Hopf algebra by [1, Proposition 6.1]. Moreover, the applications section of [2] notes that in this situation the packed algebra is also Frobenius.

For finite-dimensional weak Hopf algebras, the existence of a single faithful integral is equivalent to the underlying algebra being Frobenius; see [3, Theorem 3.16], as recalled in the concluding discussion of [14]. Hence $\widehat{\mathcal{H}}$ admits a single faithful integral. Therefore

$$N(\widehat{\mathcal{H}}) = 1.$$

$\square$

Remark 4.5. Buckley, Fieremans, Vasilakopoulou and Vercruysse also observe that, when the object set is infinite, the same packing construction yields a weak multiplier Hopf algebra which is Frobenius as an algebra. At present this does not by itself imply $N(A) = 1$, because the weak multiplier setting still leaves open whether Frobenius plus sufficiently many integrals forces the existence of a single faithful integral. Nevertheless, Theorem 4.1 shows that the most basic infinite-object examples may still satisfy $N(A) = 1$.

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